

SE404.3– Agent Based Systems

Sample Paper

Question 1: Write the correct response in your answer booklet.

[15 marks]

1. What are the two core ideas of the ABM Design Principle?
 - A. Focus on the question at hand and realism
 - B. Simplicity and Focus on the Question at hand
 - C. Simplicity and realism
 - D. Realism and any elements desired by the stakeholders
2. What are the three main components of an ABM?
 - A. Behaviors, GIS, Spatial Data
 - B. Networks, Interactions, Environments
 - C. Agents, Networks, Behaviors
 - D. Agents, Environments, Interactions
3. Why is representation in the context of ABM important?
 - A. We must combat taxation without representation.
 - B. No one understands math.
 - C. Representative agents are the key to ABM.
 - D. A good representation helps us better understand the phenomenon we are studying.
4. What are the three tabs of NetLogo?
 - A. Interface, Documents, and Procedures
 - B. Interface, Info, and Code
 - C. User, Reader, and Coder
 - D. Bar Tab, Food Tab, and Drinks Tab
5. What two buttons does almost every NetLogo model have?
 - A. Setup and Go
 - B. Create and Run
 - C. Build World and Build Turtles
 - D. Make Pink Turtles and Party
6. What are the basic mobile agents in NetLogo called?
 - A. Turtles
 - B. Agents
 - C. Humans
 - D. People
7. What is the command to clear the current model of its state?
 - A. die
 - B. destroy
 - C. clear-all
 - D. clear-drawing

8. How do you start a line that will define a procedure? And what word do you use to indicate you are done with the procedure?
- A. to and done
 - B. two spaces and end
 - C. procedure and end
 - D. to and end
9. What is the NetLogo reporter that takes an agentset and returns the members of that agentset (as an agentset) that meet a set of criteria?
- A. of
 - B. with
 - C. turtle-set
 - D. patch-set
10. What is the NetLogo command to create different types of agents?
- A. type
 - B. species
 - C. breed
 - D. set
11. Analysis of agent-based models is about understanding the relationship of _____ to _____?
- A. dependent variables, outputs
 - B. inputs, independent variables
 - C. inputs, outputs
 - D. agents, parameters
12. Agent-based modeling is not useful for modeling the behavior of a particular individual because:
- A. It only models billions of people.
 - B. Agents cannot have names.
 - C. One individual person's behavior may be too specific to capture in an agent-based model.
 - D. Agents can only be companies.
13. Verification involves comparing the _____ to the _____.
- A. real world, conceptual model
 - B. implemented model, conceptual model
 - C. implemented model, real world
 - D. my model, your model
14. Validation is the comparison of the _____ to the _____.
- A. implemented model, real world
 - B. real world, conceptual model
 - C. conceptual model, implemented model
 - D. first model, second model
15. What is the best level of representation to create a model of wolves and sheep?
- A. Each agent should represent exactly one wolf or exactly one sheep.
 - B. It depends on the question you are trying to answer.
 - C. Each agent should represent a flock of sheep or pack of wolves.
 - D. Each agent should represent one wolf and one sheep.

Question -2 (Insects moving in a grass field)

[25 marks]

- 1) Explain the basic model rules used in modeling insects moving in a grass field (8 marks)
- 2) Describe an alternative way of effectively displaying the growth of grass? Note: code is not required. (5 marks)
- 3) Imagine that group of bats are now feasting on the insects. Explain how you will model this scenario and how you will monitor the current population of each species? (5 marks)
- 4) Now suppose each bat will actively and continuously chase one insect for some time. Discuss the criteria you will use to initially lock what insect to chase and then, how you will make the bat continuously chase it. Also describe the criteria to model the death of insects now? (7 marks)

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Question-3 (Termites)

[25 marks]

Answer the following questions by referring to the NetLogo code given below.

- 1) State any assumption made in modeling the Termites model. (4 marks)
- 2) Describe the wiggle procedure (6 marks)
- 3) Describe put-down-chip turtle procedure (7 marks)
- 4) Explain the code given below with a brief introduction to the model. (8 marks)

```
to setup
  clear-all
  set-default-shape turtles "bug"
  ask patches
  [ if random-float 100 < density
    [ set pcolor yellow ] ]
  create-turtles number [
    set color white
    setxy random-pxcor random-ycor
    set size 5
  ]
end

to go
  search-for-chip
  find-new-pile
  put-down-chip
end

to search-for-chip
  ifelse pcolor = yellow
  [ set pcolor black
    set color orange
    fd 20 ]
  [ wiggle
    search-for-chip ]
end
```

```
to find-new-pile
  if pcolor != yellow
  [ wiggle
    find-new-pile ]
end

to put-down-chip
  ifelse pcolor = black
  [ set pcolor yellow
    set color white
    get-away ]
  [ rt random 360
    fd 1
    put-down-chip ]
end

to get-away
  rt random 360
  fd 20
  if pcolor != black
  [ get-away ]
end

to wiggle
  fd 1
  rt random 50
  lt random 50
end
```

Question-4 (Mousetraps)

[35 marks]

- 1) State an assumption made in the model implementation below. (5 marks)
- 2) Describe in general, what you will observe if you are to run the below program. (5 marks)
- 3) Even if a sustained chain reaction occurs, not every mousetrap gets triggered. Why? (5 marks)
- 4) What will be the shape if "Balls in the air" is plotted? Why do you think it's shaped that way? (5 marks)
- 5) Analyze the given code and explain it in a detailed manner. You need to include a brief introduction to the model in your explanation. (15 marks)

```
globals [traps-triggered]

to setup
  clear-all
  set traps-triggered 0
  ask patches
    [ set pcolor blue + 3 ]
  set-default-shape turtles "circle"
  create-turtles 1
    [ set color white
      set size 1.5 ]
  reset-ticks
end

to go
  if not any? turtles
    [ stop ]
  ask turtles
    [
      ifelse pcolor = red
        [ die ]
        [ set pcolor red
          set traps-triggered traps-triggered + 1
          hatch 1
            [ move ]
          move ] ]
  tick
end

to move
  rt random-float 360
  fd random-float max-distance
end
```